

CHA LIN, Ph.D.

Computational Biomedical Imaging | Medical AI | Multimodal Neuroimaging and Biomarkers

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RESEARCH PROFILE

Computational biomedical imaging researcher with a Ph.D. in Bioinformatics (awarded June 2026) and experience developing AI-assisted, reproducible pipelines for large-scale medical imaging and multimodal clinical data. Research spans diffusion MRI, resting-state fMRI, structural MRI, PET, cognitive and plasma biomarkers, genomics, and transcriptomics across UK Biobank, ADNI, OASIS, PPMI, and EPAD. Interested in clinically meaningful, trustworthy, and generalizable medical imaging AI for quantitative phenotyping, early disease detection, and patient-level prediction.

CORE EXPERTISE

Programming & machine learning	Python, R, MATLAB, classification, feature extraction, multimodal learning, representation learning, model evaluation
Medical imaging & quantitative analysis	FSL, SPM, FreeSurfer, ANTs, AFNI, MRtrix3, CONN; structural MRI, diffusion MRI/DTI, resting-state fMRI, PET; automated localization, image-derived phenotyping, network analysis, quality control
Multimodal modeling & validation	Clinical and cognitive outcomes, plasma biomarkers, genomic and transcriptomic data; GLM, mixed-effects and longitudinal models, survival analysis, multi-site harmonization, cross-cohort validation, interpretability and generalization

RESEARCH EXPERIENCE

2023-June 2026	AI-assisted quantitative diffusion MRI for population-scale phenotyping <i>First-author project; EBioMedicine (2026)</i> - Developed a reproducible workflow combining CNN- and YOLO-based ROI localization with standardized diffusion MRI preprocessing and rigorous quality control. - Processed imaging data from more than 60,000 UK Biobank participants and validated the workflow across PPMI, ADNI, OASIS, EPAD, and additional cohorts. - Integrated imaging-derived phenotypes with clinical, genetic, transcriptomic, molecular, vascular, and environmental information to support robust biological interpretation.
2023-2025	Functional-connectivity modeling and multimodal integration in Parkinson's disease <i>Neuroimaging lead for the imaging component; EBioMedicine (2025)</i> - Implemented resting-state fMRI preprocessing and denoising in CONN and performed Yeo-17 network seed-to-voxel analyses with confound-controlled weighted GLMs. - Supported longitudinal modeling of disease progression and integrated imaging findings with genomic and AMP-PD single-nucleus RNA-sequencing data. - Coordinated imaging outputs, quality-control decisions, and reproducible analysis across a multidisciplinary collaboration.
2022-2023	Multimodal prediction of mild cognitive impairment due to Alzheimer's disease <i>Co-first author; Journal of Alzheimer's Disease (2023)</i> - Integrated online cognitive testing with plasma p-tau181, Aβ42/40, structural MRI, and PET measures. - Trained and evaluated interpretable machine-learning classifiers (AUC up to 0.975) and related model outputs to clinically meaningful multimodal biomarkers.
2019-2022	Cancer bioinformatics and molecular oncology <i>M.S. research; Molecular Cancer Research Center, Sun Yat-sen University</i> - Analyzed TCGA and CGGA transcriptomic data to characterize glioma heterogeneity, immune-related signatures, and prognosis using differential expression, Cox regression, immune-cell deconvolution, and survival modeling. - Contributed to a study of ALG3, ferroptosis, and cancer immunotherapy, integrating molecular evidence with translational interpretation.

EDUCATION

2022-June 2026	Ph.D. in Bioinformatics, Zhongshan School of Medicine, Sun Yat-sen University, Shenzhen, China (awarded June 2026)
2019-2022	M.S. in Cell Biology, Zhongshan School of Medicine, Sun Yat-sen University, Guangzhou, China
2015-2019	B.S. in Bioengineering, School of Life Sciences, South China Normal University, Guangzhou, China

PEER-REVIEWED PUBLICATIONS

- **Lin, C.#**, Wu, H.#, Xian, W., et al. (2026). Deep learning enhanced ALPS reveals genetic and environmental factors of brain glymphatic function. *EBioMedicine*, 124, 106133.
- Wang, J., Chen, X., Du, W., **Lin, C.**, et al. (2025). Impact of Y chromosome loss on the risk of Parkinson's disease and progression. *EBioMedicine*, 117, 105769.
- Chen, W.#, **Lin, C.#**, Su, F., et al. (2023). Early diagnosis of mild cognitive impairment due to Alzheimer's disease using a composite of MemTrax and blood biomarkers. *Journal of Alzheimer's Disease*, 94(3), 1093–1103.
- Liu, P., **Lin, C.**, et al. (2022). Inhibition of ALG3 stimulates cancer cell immunogenic ferroptosis to potentiate immunotherapy. *Cellular and Molecular Life Sciences*, 79(7), 352.
- **Lin, C.**, Chen, J., Su, Z., et al. (2021). A calcium-related immune signature in prognosis prediction of patients with glioma. *Frontiers in Cell and Developmental Biology*, 9, 723103.

Equal contribution

CONFERENCE PRESENTATIONS

- Lin, C., et al. (May 2025). Deep learning enhanced neuroimaging analysis reveals risk factors linked to brain glymphatic function. Oral and poster presentation, 14th National Conference on Bioinformatics and Systems Biology.
- Pei, Z., et al. (December 2023). Early detection of mild cognitive impairment using combinations of an online cognitive assessment and blood biomarkers. Poster, Alzheimer's & Dementia Conference.

HONORS & AWARDS

- First-class Doctoral Fellowship, Sun Yat-sen University (2022–2026)
- Outstanding Graduate, Sun Yat-sen University (2022)
- National Scholarship, P.R. China (2021)

SELECTED MANUSCRIPTS IN PREPARATION

- Lin, L., Zhang, X., Chen, D., Du, W., Chen, X., Su, F., Zhang, Y., Yang, H., Lin, C., et al. Layer 5a transcriptional dysregulation links somatic CAG expansion to cortical network disintegration in Huntington's disease.
- Chen, X., Du, W., Lin, C., Liu, G. APOE ϵ 4-related brain alterations in Parkinson's disease: multimodal neuroimaging and molecular profiling.